

# **SISTEM OPERASI**

## **Laporan Tugas Praktikum 3**

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Anggota:

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PROGRAM STUDI TEKNIK INFORMATIKA

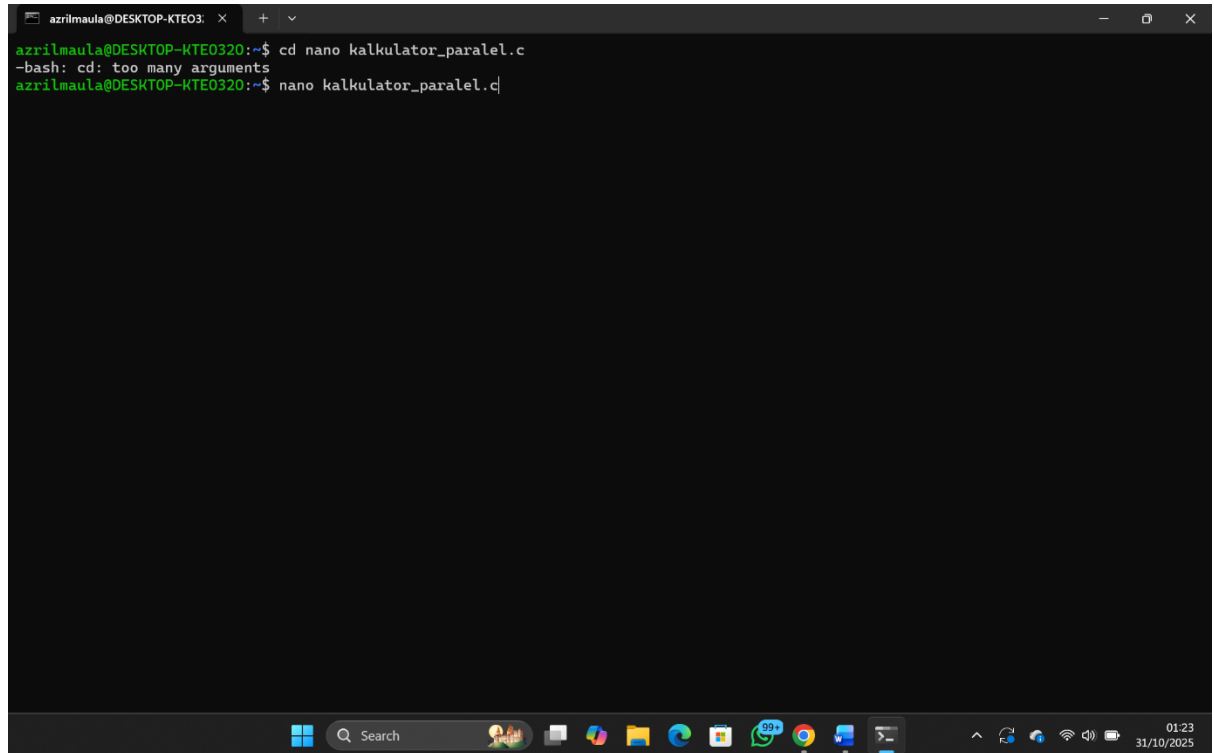
FAKULTAS TEKNIK

UNIVERSITAS MUHAMMADIYAH JEMBER

2025

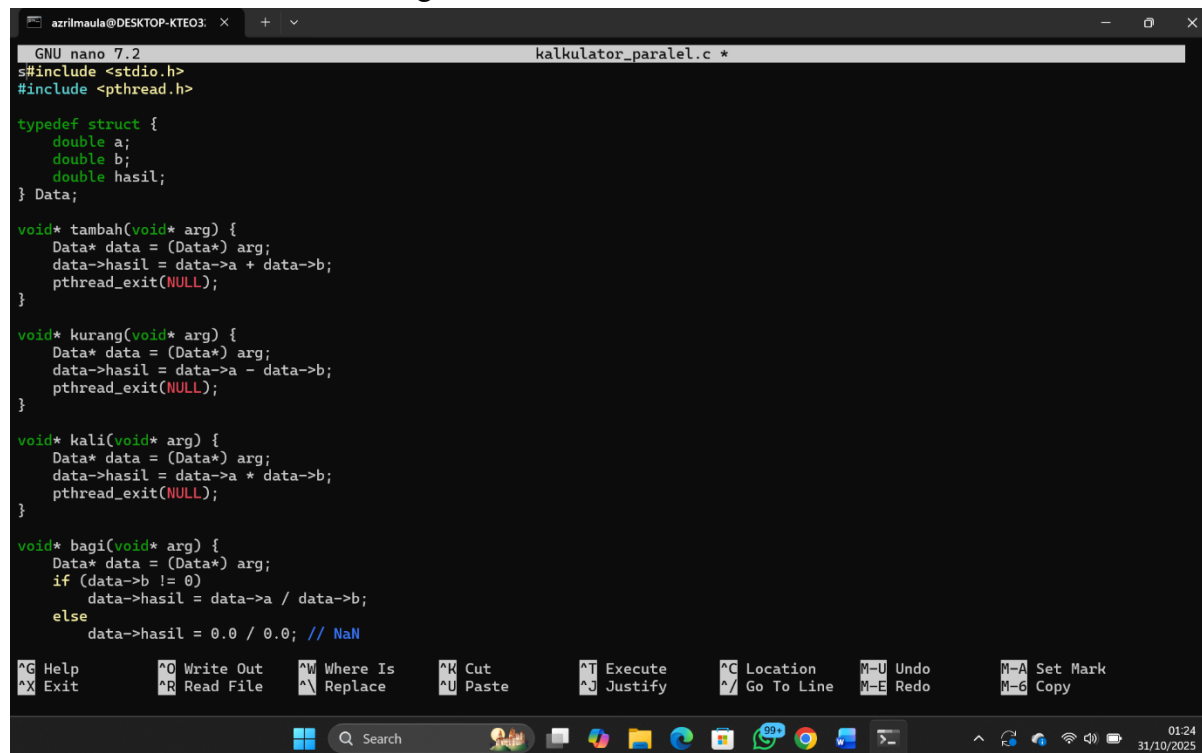
## Tugas 1 Cara Membuat Kalkulator menggunakan Bahasa c

- Pertama kita akan bikin dulu file nya dan untuk masuk ke file nya pake nano contoh seperti ini



```
azrilmaula@DESKTOP-KTE03: ~$ cd nano kalkulator_paralel.c
-bash: cd: too many arguments
azrilmaula@DESKTOP-KTE03: ~$ nano kalkulator_paralel.c
```

- Lalu setelah itu akan otomatis muncul ke dalam file tsb dan masukka beberapa source code untuk agar file kalkulator berhasil contoh :



```
GNU nano 7.2 kalkulator_paralel.c *
#include <stdio.h>
#include <pthread.h>

typedef struct {
    double a;
    double b;
    double hasil;
} Data;

void* tambah(void* arg) {
    Data* data = (Data*) arg;
    data->hasil = data->a + data->b;
    pthread_exit(NULL);
}

void* kurang(void* arg) {
    Data* data = (Data*) arg;
    data->hasil = data->a - data->b;
    pthread_exit(NULL);
}

void* kali(void* arg) {
    Data* data = (Data*) arg;
    data->hasil = data->a * data->b;
    pthread_exit(NULL);
}

void* bagi(void* arg) {
    Data* data = (Data*) arg;
    if (data->b != 0)
        data->hasil = data->a / data->b;
    else
        data->hasil = 0.0 / 0.0; // NaN
}

^G Help      ^O Write Out ^M Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo    M-A Set Mark
^X Exit      ^R Read File ^_ Replace   ^U Paste     ^J Justify   ^G Go To Line M-E Redo    M-G Copy
```

- Lalu setelah itu compile kode tsb menggunakan gcc seperti ini

```
azrilmaula@DESKTOP-KTE0320:~$ cd nano kalkulator_paralel.c
-bash: cd: too many arguments
azrilmaula@DESKTOP-KTE0320:~$ nano kalkulator_paralel.c
azrilmaula@DESKTOP-KTE0320:~$ gcc kalkulator_paralel.c -o kalkulator_paralel -lpthread
kalkulator_paralel.c:1:2: error: stray '#' in program
1 | #include <stdio.h>
  | ^
kalkulator_paralel.c:1:1: error: unknown type name 's'
1 | #include <stdio.h>
  | ^
kalkulator_paralel.c:1:11: error: expected '=', ',', ';', 'asm' or '__attribute__' before '<' token
1 | #include <stdio.h>
  | ^
kalkulator_paralel.c: In function 'main':
kalkulator_paralel.c:41:5: warning: implicit declaration of function 'printf' [-Wimplicit-function-declaration]
41 |     printf("Masukkan angka pertama: ");
    |     ^~~~~~
kalkulator_paralel.c:3:1: note: include '<stdio.h>' or provide a declaration of 'printf'
2 | #include <pthread.h>
+++ |+#include <stdio.h>
3 |
kalkulator_paralel.c:41:5: warning: incompatible implicit declaration of built-in function 'printf' [-Wbuiltin-declaration-mismatch]
41 |     printf("Masukkan angka pertama: ");
    |     ^~~~~~
kalkulator_paralel.c:41:5: note: include '<stdio.h>' or provide a declaration of 'printf'
kalkulator_paralel.c:42:5: warning: implicit declaration of function 'scanf' [-Wimplicit-function-declaration]
42 |     scanf("%lf", &a);
    |     ^~~~~~
kalkulator_paralel.c:42:5: note: include '<stdio.h>' or provide a declaration of 'scanf'
kalkulator_paralel.c:42:5: warning: incompatible implicit declaration of built-in function 'scanf' [-Wbuiltin-declaration-mismatch]
kalkulator_paralel.c:42:5: note: include '<stdio.h>' or provide a declaration of 'scanf'
azrilmaula@DESKTOP-KTE0320:~$ ./kalkulator_paralel
-bash: ./kalkulator_paralel: No such file or directory
azrilmaula@DESKTOP-KTE0320:~$ nano kalkulator_paralel.c
azrilmaula@DESKTOP-KTE0320:~$ gcc kalkulator_paralel.c -o kalkulator_paralel -lpthread
azrilmaula@DESKTOP-KTE0320:~$
```

- Dan lalu jalankan file nya dengan mengetik begini

```
azrilmaula@DESKTOP-KTE0320:~$ cd nano kalkulator_paralel.c
-bash: cd: too many arguments
azrilmaula@DESKTOP-KTE0320:~$ nano kalkulator_paralel.c
azrilmaula@DESKTOP-KTE0320:~$ gcc kalkulator_paralel.c -o kalkulator_paralel -lpthread
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  | ^
kalkulator_paralel.c:1:1: error: unknown type name 's'
1 | #include <stdio.h>
  | ^
kalkulator_paralel.c:1:11: error: expected '=', ',', ';', 'asm' or '__attribute__' before '<' token
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  | ^
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41 |     printf("Masukkan angka pertama: ");
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kalkulator_paralel.c:41:5: note: include '<stdio.h>' or provide a declaration of 'printf'
kalkulator_paralel.c:42:5: warning: implicit declaration of function 'scanf' [-Wimplicit-function-declaration]
42 |     scanf("%lf", &a);
    |     ^~~~~~
kalkulator_paralel.c:42:5: note: include '<stdio.h>' or provide a declaration of 'scanf'
kalkulator_paralel.c:42:5: warning: incompatible implicit declaration of built-in function 'scanf' [-Wbuiltin-declaration-mismatch]
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azrilmaula@DESKTOP-KTE0320:~$ ./kalkulator_paralel
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azrilmaula@DESKTOP-KTE0320:~$ nano kalkulator_paralel.c
azrilmaula@DESKTOP-KTE0320:~$ gcc kalkulator_paralel.c -o kalkulator_paralel -lpthread
azrilmaula@DESKTOP-KTE0320:~$ ./kalkulator_paralel
Masukkan angka pertama: |
```

- Lalu kita bisa memasukkan angka terserah dan lihat hasilnya  
Contoh sederhana:

```
azrilmaula@DESKTOP-KTE03: x + v
1 | $#include <stdio.h>
kalkulator_paralel.c:1:11: error: expected '=', ',', ';', 'asm' or '__attribute__' before '<' token
1 | $#include <stdio.h>
kalkulator_paralel.c: In function 'main':
kalkulator_paralel.c:41:5: warning: implicit declaration of function 'printf' [-Wimplicit-function-declaration]
41 | printf("Masukkan angka pertama: ");
kalkulator_paralel.c:3:1: note: include '<stdio.h>' or provide a declaration of 'printf'
+++ | #include <pthread.h>
3 |
kalkulator_paralel.c:41:5: warning: incompatible implicit declaration of built-in function 'printf' [-Wbuiltin-declaration-mismatch]
41 | printf("Masukkan angka pertama: ");
kalkulator_paralel.c:41:5: note: include '<stdio.h>' or provide a declaration of 'printf'
kalkulator_paralel.c:42:5: warning: implicit declaration of function 'scanf' [-Wimplicit-function-declaration]
42 | scanf("%lf", &a);
kalkulator_paralel.c:42:5: note: include '<stdio.h>' or provide a declaration of 'scanf'
kalkulator_paralel.c:42:5: note: include '<stdio.h>' or provide a declaration of 'scanf'
azrilmaula@DESKTOP-KTE0320:~$ ./kalkulator_paralel
-bash: ./kalkulator_paralel: No such file or directory
azrilmaula@DESKTOP-KTE0320:~$ nano kalkulator_paralel.c
azrilmaula@DESKTOP-KTE0320:~$ gcc kalkulator_paralel.c -o kalkulator_paralel -lpthread
azrilmaula@DESKTOP-KTE0320:~$ ./kalkulator_paralel
Masukkan angka pertama: 1
Masukkan angka kedua: 2

=== Hasil Perhitungan ===
Penjumlahan: 3.00
Pengurangan: -1.00
Perkalian : 2.00
Pembagian : 0.50
azrilmaula@DESKTOP-KTE0320:~$ |
```

## Tugas 2 membuat File Prossecing

-Pertama sama seperti biasa buat file hitung\_kata\_thread.c

```
azrilmaula@DESKTOP-KTE03: x + v
azrilmaula@DESKTOP-KTE0320:~$ nano hitung_kata_thread.c
```

- Lalu membuat kode seperti ini

```
GNU nano 7.2 hitung_kata_thread.c *
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <string.h>
#include <time.h>

#define MAX_SIZE 1000000

typedef struct {
    char *text;
    long start;
    long end;
    int count;
} ThreadData;

int is_word_char(char c) {
    return (c != ' ' && c != '\n' && c != '\t' && c != '\0');
}

// Fungsi untuk menghitung jumlah kata pada bagian tertentu
void* count_words(void* arg) {
    ThreadData* data = (ThreadData*)arg;
    int in_word = 0;
    int count = 0;

    for (long i = data->start; i < data->end; i++) {
        if (is_word_char(data->text[i])) {
            if (!in_word) {
                count++;
                in_word = 1;
            }
        } else {
            in_word = 0;
        }
    }
}
```

-lalu compile file tersebut

```
azrilmaula@DESKTOP-KTE0320:~$ nano hitung_kata_thread.c
azrilmaula@DESKTOP-KTE0320:~$ gcc hitung_kata_thread.c -o hitung_kata_thread -lpthread
azrilmaula@DESKTOP-KTE0320:~$
```

- Lalu jalankan dan coba contoh :

```
azrilmaula@DESKTOP-KTE03: ~$ nano hitung_kata_thread.c
azrilmaula@DESKTOP-KTE03: ~$ gcc hitung_kata_thread.c -o hitung_kata_thread -lpthread
azrilmaula@DESKTOP-KTE03: ~$ ./hitung_kata_thread
Masukkan nama file teks: contoh.txt

=== HASIL ===
Total kata (Single-thread): 8
Waktu eksekusi (Single-thread): 0.000013 detik
Total kata (Multi-thread): 9
Waktu eksekusi (Multi-thread): 0.002260 detik
azrilmaula@DESKTOP-KTE03: ~$
```

### Tugas 3 Membuat Producer-Consumer

- Bikin seperti biasa file nya dengan nama producer\_consumer.c

```
azrilmaula@DESKTOP-KTE03: ~$ nano producer_consumer.c
```

- lalu Masukkan Kode nya sperti ini

```
GNU nano 7.2 producer_consumer.c *
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

#define BUFFER_SIZE 10

int buffer[BUFFER_SIZE];
int count = 0; // jumlah item di buffer
int in = 0; // index tempat producer menulis
int out = 0; // index tempat consumer membaca

pthread_mutex_t mutex;
pthread_cond_t buffer_not_full;
pthread_cond_t buffer_not_empty;

// Fungsi untuk menghasilkan angka random dan menambahkannya ke buffer
void* producer(void* arg) {
    while (1) {
        int item = rand() % 100; // angka random 0-99

        pthread_mutex_lock(&mutex);

        // Tunggu jika buffer penuh
        while (count == BUFFER_SIZE)
            pthread_cond_wait(&buffer_not_full, &mutex);

        // Masukkan item ke buffer
        buffer[in] = item;
        in = (in + 1) % BUFFER_SIZE;
        count++;

        printf("Producer menghasilkan: %d | Jumlah di buffer: %d\n", item, count);
    }
}
```

-Lalu compile file tsb menggunakan gcc

```
azrilmaula@DESKTOP-KTE03: ~$ nano producer_consumer.c
azrilmaula@DESKTOP-KTE03: ~$ gcc producer_consumer.c -o producer_consumer -lpthread
```

- Lalu jalankan file nya dan hasilnya begini

```
azrilmaula@DESKTOP-KTE0320: ~$ nano producer_consumer.c
azrilmaula@DESKTOP-KTE0320: ~$ gcc producer_consumer.c -o producer_consumer -lpthread
azrilmaula@DESKTOP-KTE0320: ~$ ./producer_consumer
Producer menghasilkan: 83 | Jumlah di buffer: 1
Consumer memproses : 83 | Sisa di buffer: 0
Producer menghasilkan: 86 | Jumlah di buffer: 1
Consumer memproses : 86 | Sisa di buffer: 0
Producer menghasilkan: 77 | Jumlah di buffer: 1
Producer menghasilkan: 15 | Jumlah di buffer: 2
Consumer memproses : 77 | Sisa di buffer: 1
Producer menghasilkan: 93 | Jumlah di buffer: 2
Producer menghasilkan: 35 | Jumlah di buffer: 3
Consumer memproses : 15 | Sisa di buffer: 2
Producer menghasilkan: 86 | Jumlah di buffer: 3
Producer menghasilkan: 92 | Jumlah di buffer: 4
Consumer memproses : 93 | Sisa di buffer: 3
Producer menghasilkan: 49 | Jumlah di buffer: 4
Producer menghasilkan: 21 | Jumlah di buffer: 5
Consumer memproses : 35 | Sisa di buffer: 4
Producer menghasilkan: 62 | Jumlah di buffer: 5
Producer menghasilkan: 27 | Jumlah di buffer: 6
Consumer memproses : 86 | Sisa di buffer: 5
Producer menghasilkan: 90 | Jumlah di buffer: 6
Producer menghasilkan: 59 | Jumlah di buffer: 7
Consumer memproses : 92 | Sisa di buffer: 6
Producer menghasilkan: 63 | Jumlah di buffer: 7
Producer menghasilkan: 26 | Jumlah di buffer: 8
Consumer memproses : 49 | Sisa di buffer: 7
Producer menghasilkan: 40 | Jumlah di buffer: 8
Producer menghasilkan: 26 | Jumlah di buffer: 9
Consumer memproses : 21 | Sisa di buffer: 8
Producer menghasilkan: 72 | Jumlah di buffer: 9
Producer menghasilkan: 36 | Jumlah di buffer: 10
Consumer memproses : 62 | Sisa di buffer: 9
Producer menghasilkan: 11 | Jumlah di buffer: 10
^C
azrilmaula@DESKTOP-KTE0320: ~$
```



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Search



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31/10/2025